Circle the correct answer for each question. If you're unsure, you can write numbers 1, 2, 3, etc. next to the choices to get partial credit.

**Question permutation table:**

| Version | | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Question | | | | | |
| 1 | → | 8 | 1 | 4 | 8 |
| 2* | → | 9 | 2 | 5 | 9 |
| 3* | → | 10 | 3 | 6 | 10 |
| 4 | → | 5 | 8 | 8 | 11 |
| 5 | → | 11 | 7 | 7 | 1 |
| 6 | → | 4 | 6 | 11 | 5 |
| 7 | → | 6 | 4 | 9 | 6 |
| 8* | → | 7 | 5 | 10 | 7 |
| 9 | → | 1 | 9 | 1 | 2 |
| 10* | → | 2 | 10 | 2 | 3 |
| 11* | → | 3 | 11 | 3 | 4 |

*: Question follows previous

Suppose a random sample of 100 students is taken at a university, asking the students how many hours they slept last night. The sample has a mean of 7.1 hours and a standard deviation of 1.2 hours. A 95% confidence interval is then constructed for the mean number of hours all students from that university slept last night. Use this to answer the following three questions:

1. Suppose, instead, that a 90% confidence interval is constructed with the data. How would the 90% confidence interval compare to your 95% confidence interval? The 90% confidence interval would be:

   (a) Wider
   (b) Narrower
   (c) Same width
   (d) Cannot be determined with the information provided

   **Correct answers:** (b)

   **Answer permutations:** (All answers permuted)

| Version | | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Answer | | | | | |
| (a) | → | (b) | (c) | (a) | (c) |
| (b) | → | (a) | (a) | (d) | (a) |
| (c) | → | (d) | (b) | (b) | (d) |
| (d) | → | (c) | (d) | (c) | (b) |

2. Suppose another random sample is taken with 150 students, and has the same sample mean and standard deviation as the original sample. How would the new 95% confidence interval compare to the original? The the new interval would be:

   (a) Wider
   (b) Narrower
   (c) Same width
   (d) Cannot be determined with the information provided

   **Correct answers:** (b)

   **Answer permutations:** (All answers permuted)

| Version | | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Answer | | | | | |
| (a) | → | (c) | (b) | (c) | (d) |
| (b) | → | (a) | (c) | (d) | (c) |
| (c) | → | (b) | (a) | (b) | (b) |
| (d) | → | (d) | (d) | (a) | (a) |

3. Suppose another sample of 100 students is taken and this time the mean is 7.1 with a standard deviation of 1.4 for the hours slept. These new statistics are used to create another 95% confidence interval for the mean hours slept. How would the new 95% confidence interval compare to the original? The new 95% confidence interval would be:

   (a) Wider
   (b) Narrower
   (c) Same width
   (d) Cannot be determined with the information provided

   **Correct answers:** (a)

   **Answer permutations:** (All answers permuted)

| Version | | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Answer | | | | | |
| (a) | → | (c) | (d) | (d) | (d) |
| (b) | → | (d) | (c) | (a) | (b) |
| (c) | → | (a) | (a) | (c) | (a) |
| (d) | → | (b) | (b) | (b) | (c) |

4. A 2012 survey of 1,254 American cell phone owners indicates that 17% of those surveyed do browsing on their phone rather than a computer or other device.
   Suppose another survey was done on 1,254 American cell phone owners, but this time, 23% of cell phone owners surveyed do browsing on their phone rather than a computer or other device. How would the 95% confidence interval from this survey compare to the original? The new survey's interval would be:

   (a) Wider
   (b) Narrower
   (c) Same width
   (d) Cannot be determined with the information provided

   **Correct answers:** (a)

   **Answer permutations:** (All answers permuted)

| Version | | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Answer | | | | | |
| (a) | → | (a) | (c) | (a) | (a) |
| (b) | → | (b) | (a) | (c) | (d) |
| (c) | → | (d) | (b) | (b) | (b) |
| (d) | → | (c) | (d) | (d) | (c) |

5. Suppose data is collected on the heights of all students at a college. Each of the 250 people taking a statistics course randomly takes a sample of 40 of these heights and constructs a 95% confidence interval for mean height of all students at the college.
   Which of the following statements about the confidence intervals is most accurate?

   (a) About 95% of the heights of all students at the college will be contained in these intervals.
   (b) About 95% of the time, a student's sample mean height will be contained in the interval the student found.
   (c) About 95% of the intervals will contain the population mean height.
   (d) About 95% of the intervals will be identical.

   **Correct answers:** (c)

   **Answer permutations:** (All answers permuted)

| Version | | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Answer | | | | | |
| (a) | → | (b) | (a) | (b) | (a) |
| (b) | → | (d) | (b) | (a) | (b) |
| (c) | → | (a) | (c) | (c) | (c) |
| (d) | → | (c) | (d) | (d) | (d) |

6. Researchers are studying a new weight-loss program. Using a sample of 9,812 volunteers with a desire to lose weight, a 95% confidence interval is constructed for the true mean amount of weight loss after six months on the program, and found to be (0.12, 0.20) pounds.
Based on this interval, does statistically significant evidence exist that the weight-loss program works?

   (a) Yes, since the entire interval is positive.
   (b) Yes, since the sample size was large.
   (c) No, since the true mean amount of weight loss after six months on the program is less than a pound.
   (d) The question cannot be answered without the p-value.

**Correct answers:** (a)
**Answer permutations:** (All answers permuted)

| | Version | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Answer | | | | | |
| (a) | → | (c) | (c) | (d) | (a) |
| (b) | → | (b) | (a) | (c) | (b) |
| (c) | → | (d) | (d) | (a) | (c) |
| (d) | → | (a) | (b) | (b) | (d) |

The following table shows the responses from a sample of 680 people in the General Social Survey to the question, "Do you sometimes drink more than you think you should?"

| Gender | Yes | No | Total |
|---|---|---|---|
| Male | 151 | 177 | 328 |
| Female | 92 | 260 | 352 |
| Total | 243 | 437 | 680 |

7. What proportion of males responded that they drink more than they should?

   (a) 151/243
   (b) 151/328
   (c) 243/680
   (d) 151/680

**Correct answers:** (b)
**Answer permutations:** (All answers permuted)

| | Version | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Answer | | | | | |
| (a) | → | (c) | (b) | (b) | (c) |
| (b) | → | (d) | (a) | (a) | (d) |
| (c) | → | (a) | (d) | (d) | (a) |
| (d) | → | (b) | (c) | (c) | (b) |

8. Does there appear to be an association between gender and whether a person responds that they drink more than they should?

   (a) Yes, since a different number of people of each gender are in the sample.
   (b) Yes, since the percent that responded that they drink more than they should differs from 50
   (c) Yes, since the proportion who responded they drink more than they should differs between males and females.
   (d) No, there does not appear to be an association between these two variables.

**Correct answers:** (c)
**Answer permutations:** (All answers permuted)

| | Version | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Answer | | | | | |
| (a) | → | (a) | (c) | (b) | (c) |
| (b) | → | (c) | (b) | (d) | (a) |
| (c) | → | (d) | (a) | (c) | (b) |
| (d) | → | (b) | (d) | (a) | (d) |

A study published in 2007 by Christopher Johnson, professor of music education and music therapy at the University of Kansas, revealed that students in elementary schools who took private music lessons scored around 20 percent higher in math scores on standardized tests, compared to students who did not take private music lessons. Use this to answer the following three questions:

9. What is the explanatory variable?

   (a) Music therapy
   (b) Whether a student took private music lessons
   (c) Math score on standardized test
   (d) Grade level

   **Correct answers:** (b)

   **Answer permutations:** (All answers permuted)

| | Version | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Answer | | | | | |
| (a) | → | (d) | (a) | (b) | (b) |
| (b) | → | (c) | (b) | (a) | (d) |
| (c) | → | (b) | (d) | (d) | (c) |
| (d) | → | (a) | (c) | (c) | (a) |

10. What is the response variable?

    (a) Music therapy
    (b) Whether a student took private music lessons
    (c) Math score on standardized test
    (d) Grade level

    **Correct answers:** (c)

    **Answer permutations:** (All answers permuted)

| | Version | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Answer | | | | | |
| (a) | → | (a) | (d) | (c) | (d) |
| (b) | → | (d) | (c) | (d) | (c) |
| (c) | → | (b) | (a) | (a) | (a) |
| (d) | → | (c) | (b) | (b) | (b) |

11. Which of the following could be a potential confounding variable?

    (a) Household income
    (b) Socio-economic status
    (c) Whether the parents hired a math tutor
    (d) All of these

    **Correct answers:** (d)

    **Answer permutations:** (All answers permuted)

| | Version | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Answer | | | | | |
| (a) | → | (a) | (d) | (a) | (a) |
| (b) | → | (c) | (a) | (c) | (c) |
| (c) | → | (d) | (b) | (d) | (b) |
| (d) | → | (b) | (c) | (b) | (d) |